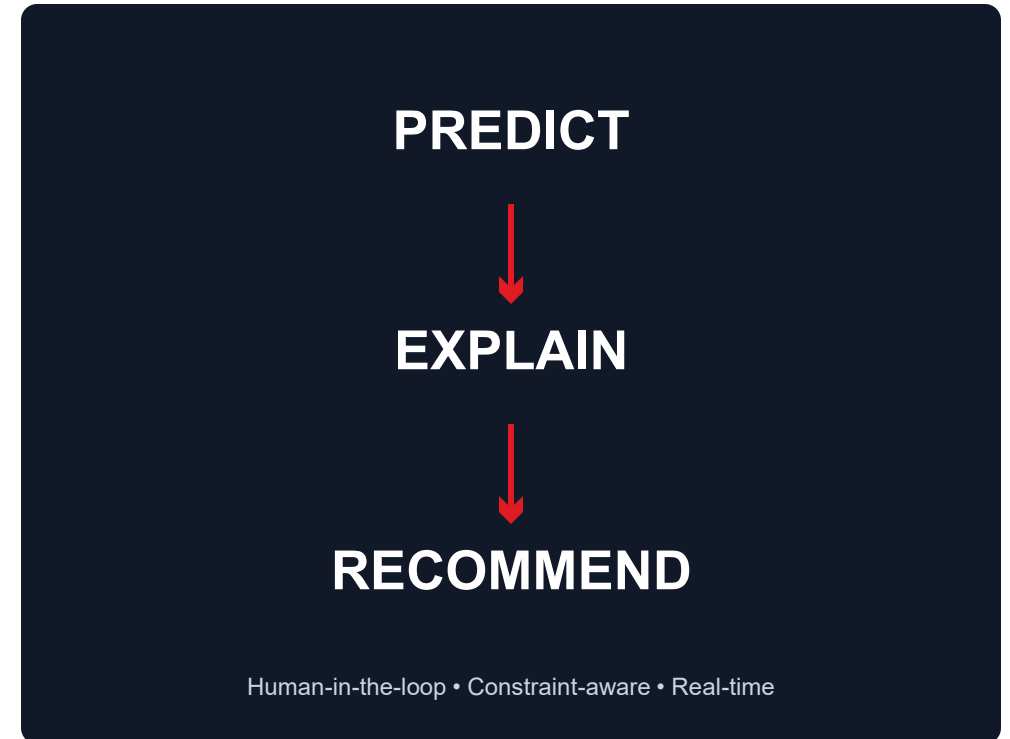


GRADESENSE

Explainable AI for safer, faster paper grade transitions

- **Honeywell Hackathon Final Submission**
- **Industrial AI | Software Prototype**
- **GradeSense Team**



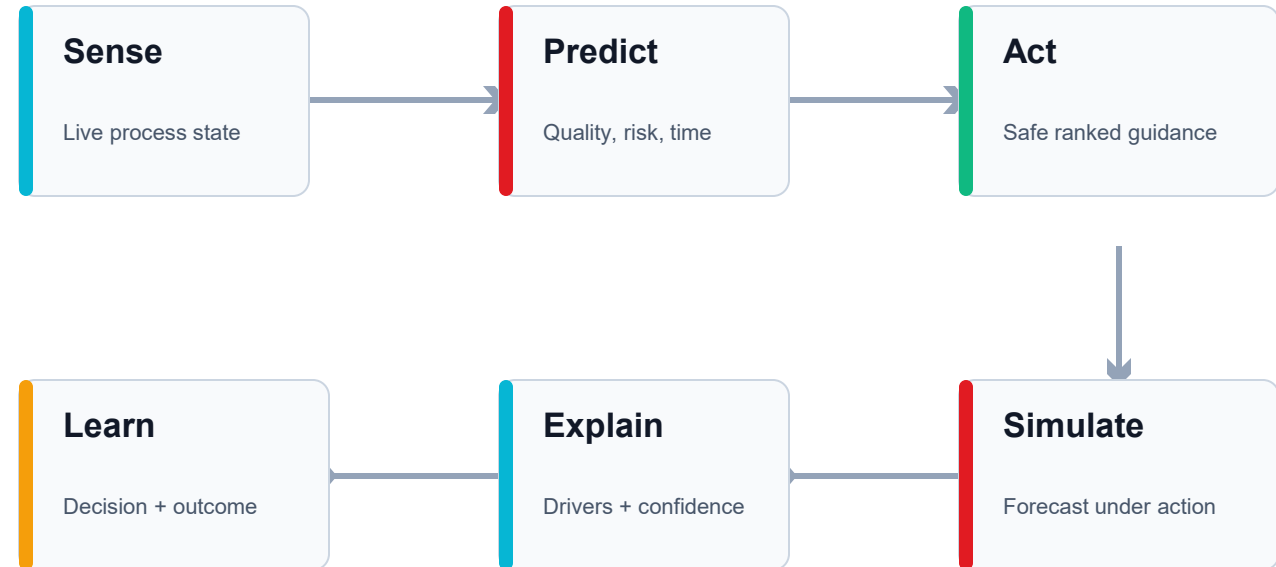
PREDICT BEFORE QUALITY DRIFTS

THE PROBLEM

Grade transitions are multivariable, time-dependent, and costly when deviation is detected late.

Operators must connect changing sensor values, future basis-weight behavior, safety constraints, and past outcomes under time pressure.

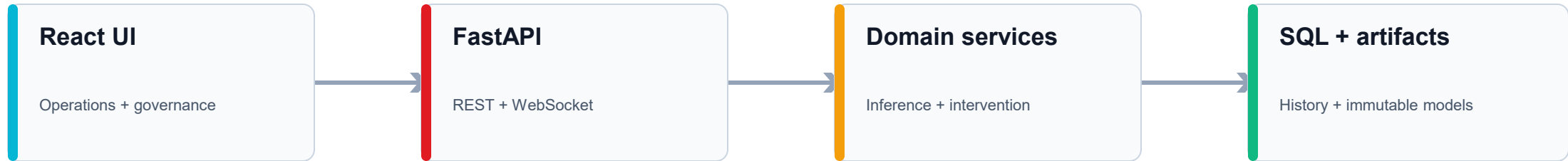
THE GRADE SENSE LOOP



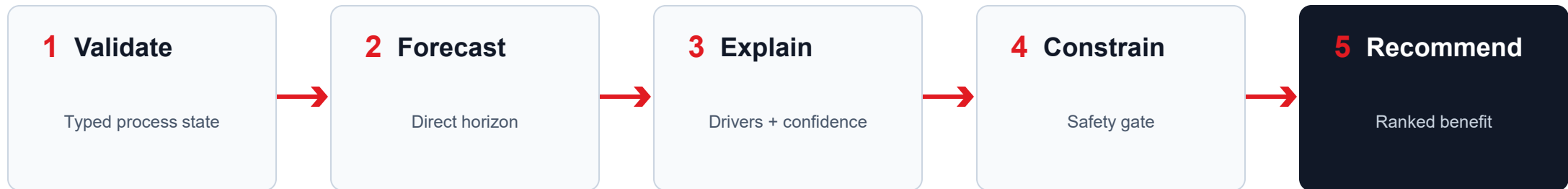
One governed system connects detection, explanation, intervention, and measurable learning.

PRODUCTION-READY AI ARCHITECTURE

SYSTEM ARCHITECTURE



AI WORKFLOW



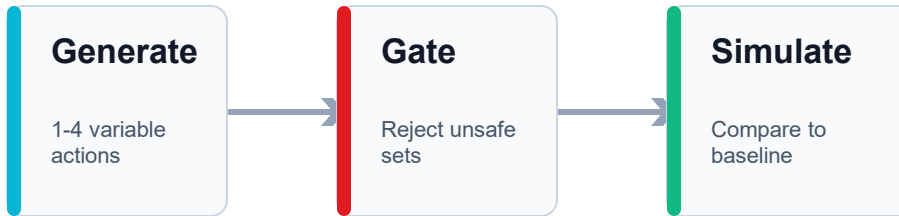
Python • FastAPI • SQLAlchemy • scikit-learn • React • TypeScript • PostgreSQL • Docker

EXPLAINABILITY, SAFETY, AND OUTCOMES

EXPLAINABLE AI



RECOMMENDATION ENGINE



**Only positive, constraint-valid improvements
are ranked for the operator.**

BUSINESS IMPACT

EARLIER

risk detection

SAFER

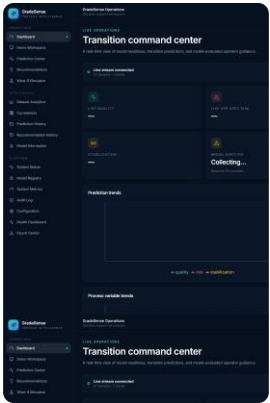
set-point guidance

TRACEABLE

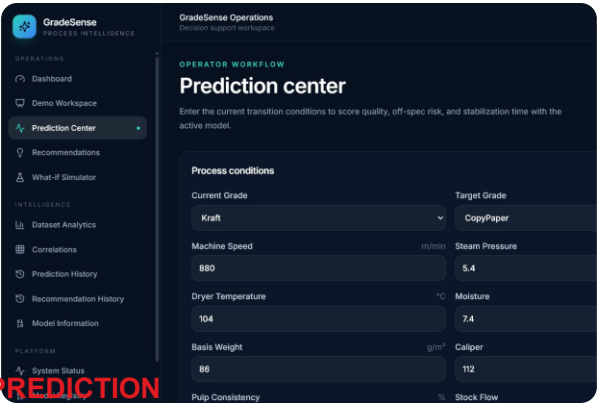
decision-to-outcome loop

Advisory by design: the operator remains in control.

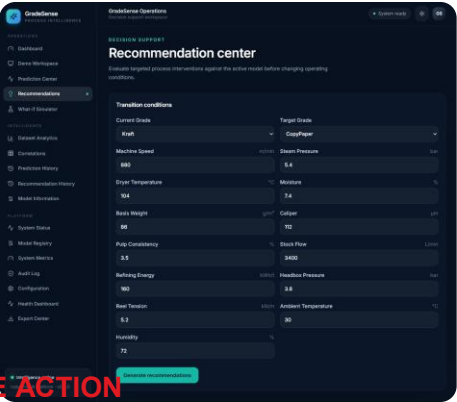
WORKING DASHBOARD: THE CLOSED LOOP



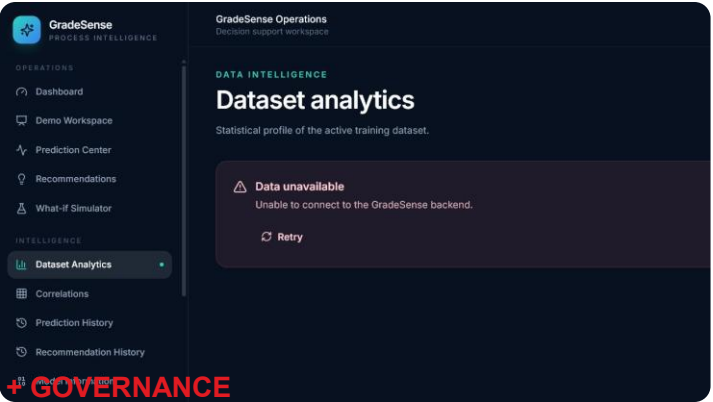
LIVE OPERATIONS



EXPLAINABLE PREDICTION



CONSTRAINT-AWARE ACTION



ANALYTICS + GOVERNANCE

DEPLOYABLE TODAY; PILOT-READY NEXT

DEPLOYMENT ARCHITECTURE



REFERENCE DELIVERY

Docker Compose • Render blueprint • Alembic migrations
• health checks • read-only model mounts

PILOT NEXT

Connect PLC/SCADA data
Calibrate on plant history
Enable SSO and governed roles
Measure off-spec and stabilization outcomes

GradeSense turns model output into safe, explainable, measurable operator action.